DGCA CPL / ATPL STUDY NOTES • RADIO NAVIGATION • CH 20

Chapter 20: Quick Reference & Index

All Key Facts, Formulae, Frequencies & Exam Tips — All Chapters

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Essential Formulae & Calculations

▶ Radio Waves

c = f × λ | c = 3×10&sup8; m/s

λ(m) = 300 / f(MHz)

f(MHz) = 300 / λ(m)

c = 162,000 NM/s

▶ LOS Range

Radar: 1.23 × ($\sqrt{h_{TX}}$ + $\sqrt{h_{RX}}$) NM

VOR/VDF: 1.25 × ($\sqrt{h_{TX}}$ + $\sqrt{h_{RX}}$) NM

(heights in feet)

▶ Radar Range & PRF

Max range (NM) = 81,000 / PRF (pps)

Max range (km) = 150,000 / PRF (pps)

PRF = 81,000 / Max range (NM)

Range (NM) = echo time (µs) / 12.36

Min range (m) = PW (µs) × 150

▶ Power vs Range

Power ∝ Range&sup4;

Double range → power × 2&sup4; = ×16

Triple range → power × 81

▶ DME

Range (NM) = (echo time µs − 50) / 12.36

Transponder offset: ±63 MHz

Delay: 50 µs

Slant error negligible: Range(NM) > 3 × Alt(000s ft)

Search: 150 ppps | Acq: 60 ppps | Track: 27 ppps

Saturation: 2700 ppps ≈ 100 aircraft

▶ ILS Height Check

H(ft) ≈ GS angle° × Range(NM) × 100 + threshold ht

Example: 3° at 4.6 NM + 50 ft = 1430 ft

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▶ GPS

4 SVs for 3D fix + time

5 SVs for RAIM detection

6 SVs for RAIM exclusion

Horizontal: ±13 m (95%) | Vertical: ±22 m (95%)

Time: ±40 ns (95%)

▶ NDB Range vs Power

P ∝ R&sup4;

Range 10 NM → 20 NM: power ×16

ADF day accuracy: ±5°

VDF Class A: ±1° | B: ±2° | C: ±5° | D: ±10°

Frequency & Emission Reference

System	Frequency / Band	Emission	Key Facts
NDB	190–1750 kHz (LF/MF)	MCW (A2A) / NON	Omni signal; ADF measures RB; BFO for NON beacons
VDF	VHF (108–174 MHz)	A3E	LOS only; aircraft needs VHF radio only
VOR	108–117.95 MHz (VHF)	A9W / A2A (ident)	Phase comparison 9960 Hz FM ref vs 30 Hz AM var; ident 1020 Hz
ILS LOC	108–111.975 MHz (VHF, odd decimals)	A9W	90 Hz + 150 Hz DDM; 40 channels
ILS G/S	329–335 MHz (UHF)	A9W	Auto-paired with LOC; range 10 NM; ±0.5°
ILS Markers	75 MHz (VHF)	A2A	OM:400Hz/blue | MM:1300Hz/amber | IM:3000Hz/white
DME	960–1215 MHz (UHF)	P0N	252 ch; ±63 MHz offset; 50µs delay; ±0.5NM/3%
SSR	1030 MHz (gnd TX) / 1090 MHz (A/C TX)	P0N	4096 codes (Mode A); 16.7M (Mode S)
AWR	9375 MHz (SHF, 3.2 cm)	P0N	Primary radar; pencil + fan beam
MLS	5031–5090.7 MHz (SHF)	—	200 ch; TRSB; ±40° az; 0.9°–20° GS
ASMI	15–17 GHz (SHF, ~2 cm)	—	60 rpm; PW 0.03µs; PRF 4000–20000 pps
GPS L1	1575.42 MHz	—	Civil C/A code; 1.023 MHz chip rate
GPS L2	1227.6 MHz	—	Military P-code; 10.23 MHz chip rate
GLONASS	~1600 MHz / ~1246 MHz	—	FDMA; 24 SVs; 65° incl; PZ90
Galileo	E1/E5a/E5b/E6 (~1.2–1.6 GHz)	—	30 SVs; 56° incl; ETRS89

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✈ ILS Categories

Category	Decision Height	RVR	Notes
CAT I	200 ft	800 m	Standard ILS
CAT II	100 ft	400 m	Autoland capable aircraft required
CAT IIIA	100 ft	200 m	Autoland; enhanced RVR system
CAT IIIB	<50 ft	50–200 m	Low visibility procedures
CAT IIIC	None	None	Theoretical; no current certification

ILS False Glideslope: 3° nominal → false slopes at 9° (3×), 15° (5×). Strongest false slope at **3× nominal**. If established on false GS, it indicates ABOVE actual GS but aircraft is too steep.

📝 Transponder Codes — Emergency Reference

🔴 7700 — EMERGENCY

General emergency; activates alert on ATC display

🔵 7600 — RADIO FAILURE

NORDO; follow lost comms procedures

🔶 7500 — HIJACK

Unlawful interference; covert squawk if possible

7000 — VFR Conspicuity (UK)

Standard VFR code in UK; Mode C if available

2000 — Inadvertent CAS Entry

Entering controlled airspace without clearance

1200 — VFR (USA)

Standard US VFR code (not used in UK)

❁ GNSS Systems — Full Comparison

Parameter	GPS	GLONASS	Galileo
SVs (nominal)	24 (+spares)	24	30
Orbital planes	6	3	3
SVs per plane	4	8	10
Altitude	20,180 km	19,099 km	23,222 km
Inclination	55°	65°	56°
Period	11 h 56 min	11 h 15 min	14 h 8 min

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Geoid	WGS84	PZ90	ETRS89
Multiple access	CDMA	FDMA	CDMA
L1 frequency	1575.42 MHz	~1602 MHz	1575.42 MHz

GPS Navigation Message Sub-frames: SF1 = clock corrections | SF2+3 = ephemeris (orbital data) | SF4 = ionospheric model, UTC | SF5 = almanac
Almanac download: 25 frames × 30 s = **12.5 minutes** (cold start)

📱 Radar System Specifications

System	Frequency	PRF	Scan Rate	Max Range	Purpose
ASR (En-route)	SHF	200–400 pps	5–6 rpm	200–300 NM	Area surveillance
TAR (Terminal)	SHF	~400 pps	15 rpm	75 NM	Terminal area
Aerodrome PAR	SHF	—	Sector	25 NM	Precision approach
ASMI	15–17 GHz	4000–20000 pps	60 rpm	4–5 NM	Surface movement
AWR	9375 MHz	variable	stabilised	150–320 NM	Weather

🌩 AWR — Colour Guide & Hazardous Shapes

Colour	Weather Mode	Map Mode
BLACK	<0.7 mm/h — nil/very light	—
GREEN	0.7–4 mm/h — light	Light return
YELLOW	4–12 mm/h — moderate	Medium return
RED	>12 mm/h — heavy	Heavy return
MAGENTA	Turbulence (severe)	—

⚠ Hazardous Return Shapes — AVOID ALL

Hook shape: tornado / severe turbulence
U-shape (hollow core): hail + severe turbulence
Finger (narrow projection): hail activity
Scalloped edges: hail at storm boundary
Shadow area: severe weather MAY be hidden beyond heavy rain
CONT mode dark bands: steepest rainfall gradient = worst hazard

AWR Tilt Rule: Tilt UP to see storm tops (severity assessment). Tilt DOWN to scan for weather at range. Use MAP mode (fan beam) up to 60–70 NM, then switch to MAN (pencil) beyond.

📡 SSR Modes — Quick Reference

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Mode	Interrogation Spacing	Information	Codes/Resolution
Mode A	P1–P3: 8 µs	Identity (squawk)	4096 codes (octal 0000–7777)
Mode C	P1–P3: 21 µs	Pressure altitude (1013.25 hPa)	100 ft increments (Gillham code)
Mode S	24-bit address	Data link; ID; alt; ELS/EHS	16,777,216 codes; 25 ft alt resolution

ELS (Elementary Surveillance)

Squawk | Callsign | Pressure altitude | Emergency

EHS (Enhanced Surveillance)

ELS + Selected altitude | Airspeed | Magnetic heading

SSR Issues: Garbling — two aircraft within 1.7 NM same direction | Fruiting — phantom targets from other interrogators (fixed by ISLS) | SLS — P2 omnidirectional pulse suppresses side-lobe replies

📌 RNAV Accuracy Standards

Type	Lateral Accuracy	% Requirement	Application
B-RNAV	±5 NM	95%	Mandatory: Eurocontrol FIR >30 pax jets
P-RNAV	±1 NM	95%	Terminal area routes
2D RNAV	Lateral only	—	VOR/DME rho/theta
3D RNAV	Lateral + Vertical	—	VNAV added
4D RNAV	Lateral + Vertical + Time	—	RTA (Required Time of Arrival)

RNAV Erratic: Most common cause = aircraft **beyond DOC / below LOS of reference VOR/DME**. Also: reference VOR cone of confusion (overhead). Phantom station itself has no broadcast — its position is computed.

💻 EFIS Display Modes

Mode	Orientation	WXR/TCAS	Bearing Pointer
Full Rose — VOR	Heading-up	No	Yes
Full Rose — ILS	Heading-up	No	No
Expanded VOR	Heading-up	Yes	Yes
Expanded ILS	Heading-up	Yes	No
MAP	Heading-up	Yes	Yes
PLAN	**True North-up**	No	No

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⚠ Error Sources — All Systems

ADF / NDB Errors

- Night effect (sky wave) — WORST
- Quadrantal error (airframe)
- Coastal refraction
- Mountain effect
- Static interference
- Station interference
- No failure warning

VOR Errors

- Site error (terrain reflections) — WORST
- Scalloping (multipath)
- Cone of confusion (>30° elevation)
- Propagation anomalies
- Airborne equipment error
- Pilot error (OBS)

GPS Errors

- Ionospheric delay (single freq)
- Tropospheric delay
- Multipath
- Receiver clock error
- Ephemeris errors
- PDOP (geometry)
- Selective Availability (off)

ILS Errors

- Multipath reflections from terrain/buildings
- False glideslopes (3×, 5×)
- Beam bends near threshold
- Back course reversal
- Runway contamination

DME Errors

- Slant range vs ground range
- Garbling (multiple aircraft)
- Transponder saturation
- RNAV cone of confusion (reference VOR)

AWR Limitations

- Cannot detect CAT
- Shadow behind heavy rain
- Attenuation in heavy precip
- Blue ice undetected
- Stabilisation failure → lopsided display

🔴 Exam Traps — Don't Mix These Up!

Topic	WRONG (common error)	CORRECT
VOR variation	At aircraft	At **station**
ADF variation	At station	At **aircraft**
VOR ident tone	1350 Hz	**1020 Hz**
DME ident tone	1020 Hz	**1350 Hz**
ILS LOC band	329–335 MHz	**108–111.975 MHz VHF**
ILS G/S band	108–118 MHz	**329–335 MHz UHF**
SSR ground TX	1090 MHz	**1030 MHz**
SSR aircraft TX	1030 MHz	**1090 MHz**
GPS SVs for RAIM detect	4 or 6	**5 SVs**
GPS SVs for RAIM exclude	5 or 7	**6 SVs**
PLAN mode orientation	Heading-up	**True North-up**

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Mode C altitude ref	QNH or local	**1013.25 hPa (always)**
MLS channels	40 or 252	**200 channels**
ILS channels	200 or 252	**40 channels**
DME channels	40 or 200	**252 channels**
GLONASS inclination	55° or 56°	**65°**
GPS inclination	63° or 65°	**55°**
Max radar range limiter	Pulse width	**PRF**
Min radar range limiter	PRF	**Pulse width**
AWR type	Secondary radar	**Primary radar**
DME/SSR type	Primary radar	**Secondary radar**
ILS false GS angle	2× or 5× nominal	**3× nominal** (strongest)
GPS altitude reference	MSL	**WGS84 ellipsoid**
7500 squawk	Emergency / Radio fail	**Hijack only**

🎓 **Mnemonics:**

ILS colours (low to high frequency for markers): **B**lue **A**mber **W**hite = **B**ig **A**pproach **W**arning (OM→MM→IM)

AWR colours: **B**lack **G**reen **Y**ellow **R**ed **M**agenta = **B**ig **G**reen **Y**achts **R**ace **M**adly

SSR emergency codes: 75=**Hi**jack / 76=R**a**dio / 77=**E**mergency = **HAE**

GPS RAIM: 4=Fix | 5=Detect | 6=Exclude = **4-5-6**

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DGCA CPL/ATPL Radio Navigation Study Notes

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